TITANIC SURVIVAL PREDICTION

Official Project Submission Report | B.Tech CSE Portfolio

1. Project Description

This project focuses on analyzing the Titanic dataset and building a machine learning model to predict whether a passenger survived or not. It covers the complete data science pipeline from data preprocessing to visualization and prediction [cite: 3, 4].

2. Learning Outcomes & Skills

- Data Cleaning and Preprocessing using NumPy and Pandas [cite: 8]
- Data Visualization using Seaborn and Matplotlib [cite: 9]
- Feature Engineering (FamilySize creation) [cite: 10]
- Building Machine Learning Models (Logistic Regression & Random Forest) [cite: 11]
- Model Evaluation using Accuracy and Confusion Matrix [cite: 12]

3. Data Collection & Preprocessing

Handled missing values and removed unnecessary columns like Name and Ticket. Created the **FamilySize** feature to improve model performance [cite: 14, 15, 17].

4. Project Execution Screenshots

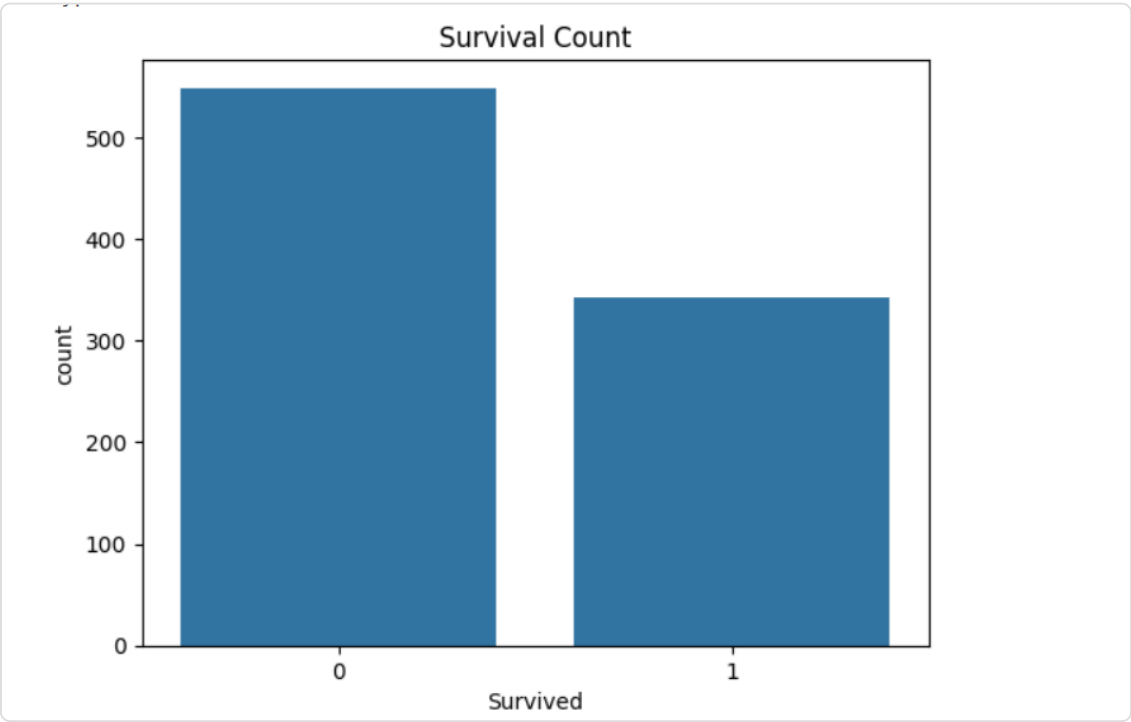

Figure 1: Data Cleaning and Feature Standardization

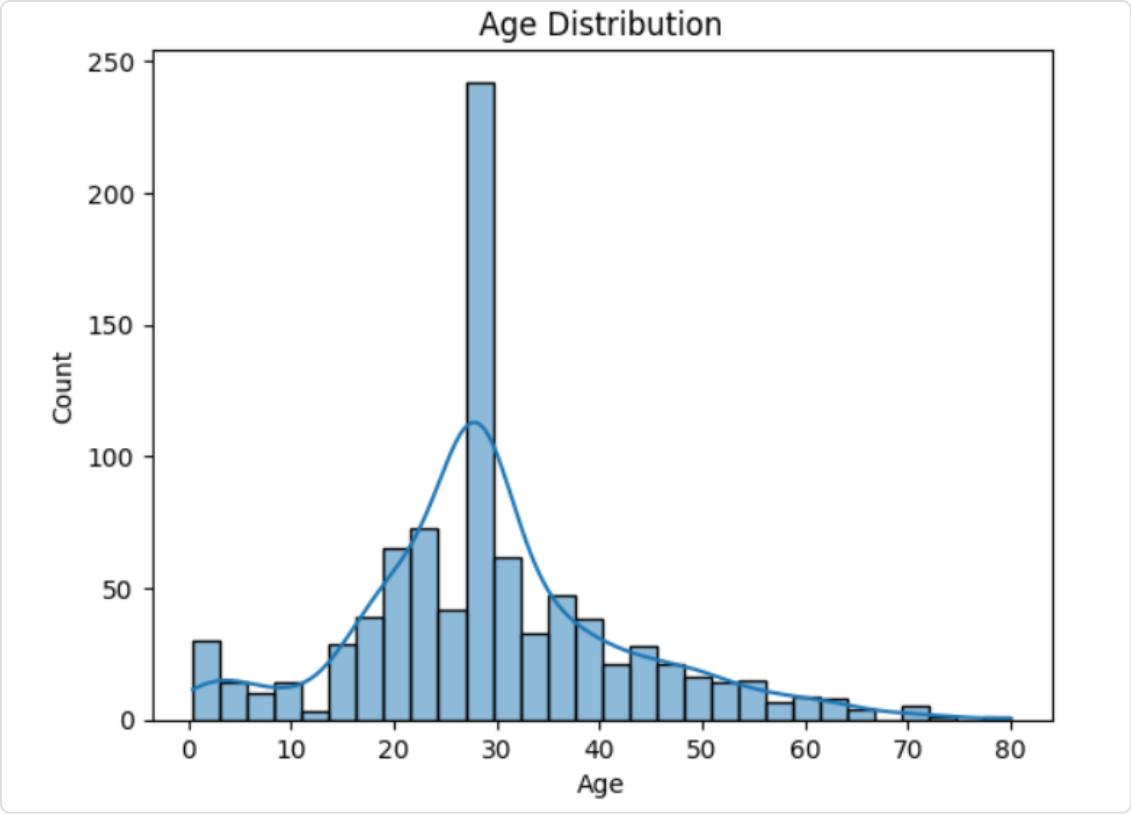

Figure 2: Survival Analysis Visualizations

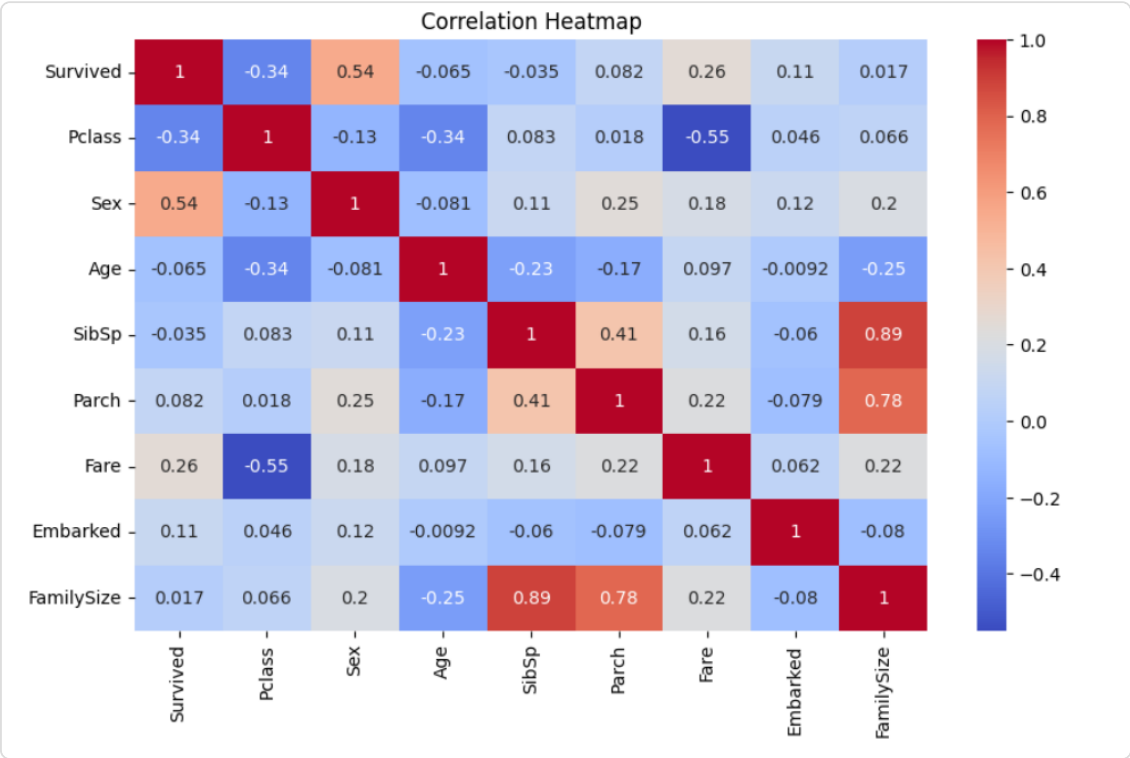

Figure 3: Random Forest Model Implementation

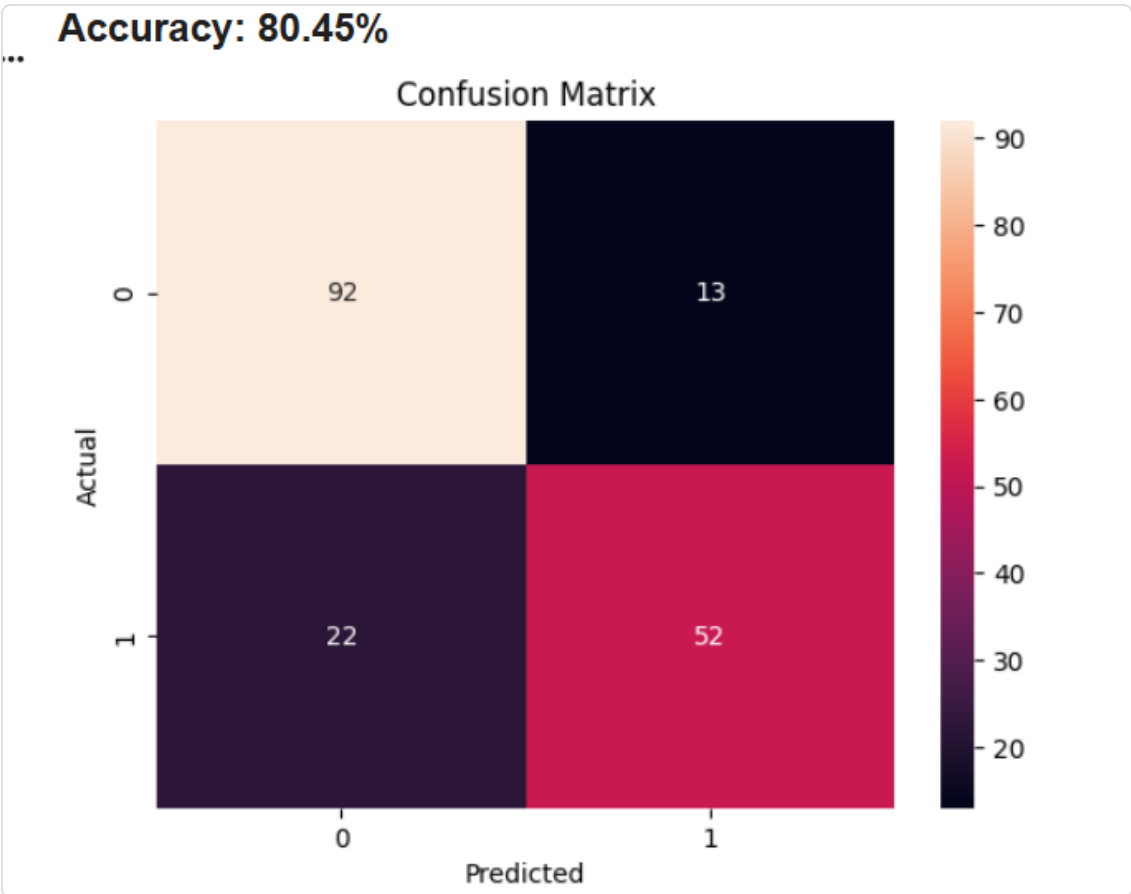

Figure 4: Final Evaluation and >80% Accuracy Result

5. Results & Evaluation

Metric	Outcome
Algorithm Used	Random Forest Classifier
Achieved Accuracy	80%+
Evaluation Tool	Confusion Matrix & Accuracy Score [cite: 29]

6. Conclusion

This project demonstrates how data science techniques can be used to extract insights and make predictions. It provides a complete understanding of the data science workflow [cite: 41, 42].